

- 7 The photograph shows a corpse lily flower (*Rafflesia hasseltii*).



(Source: © paul kennedy/Alamy Stock Photo)

The corpse lily plant is found in the rainforests of Sumatra.

The flowers can be 100 cm in diameter and are the largest flowers found on Earth.

- (a) The smallest flower on Earth is produced by *Wolffia globosa*.

The flower of *W. globosa* measures $50\text{ }\mu\text{m}$ in diameter.

Which is the ratio for the diameters of the flowers of *R. hasseltii* to *W. globosa*?

(1)

- ☐ A 20:1
- ☐ B 200:1
- ☐ C 2000:1
- ☐ D 20 000:1



- (b) The corpse lily plant is unusual because none of its cells contains chloroplasts.

Most of the corpse lily plant grows inside the tissues of a rainforest vine plant, including inside the tissues involved in the translocation of organic solutes.

Only the corpse lily flower can be seen growing outside of the vine.

The corpse lily plant obtains organic solutes such as sucrose from one of the tissues in the vine plant.

- (i) State what is meant by the term **tissue**.

(1)

- (ii) Suggest how the corpse lily plant obtains organic solutes, such as sucrose, from one of the tissues in the vine plant.

(3)

- (c) The wound healing properties of an extract made from corpse lily plants were investigated.

Treatments were tested on four groups of six rats with identical skin wounds.

The treatments were applied to the wounds twice a day.

The time taken for the wounds to heal was recorded.

The table shows the treatment applied to each group and the results of this investigation.

Rat group	Treatment	Mean healing time / days	Standard deviation
1	gel + no extract	21.67	± 0.48
2	gel + 5% extract	15.83	± 0.33
3	gel + 10% extract	14.67	± 0.56
4	gel + current wound treatment	12.33	± 0.31

- (i) Explain the purpose of the gel used in these treatments.

(3)



(ii) Comment on the results of this investigation.

(4)

(Total for Question 7 = 12 marks)

